PHARM 201 Lecture Notes: Pharmacokinetics and Antihypertensives

Pharmacokinetics and Pharmacodynamics

Pharmacokinetics describes what the body does to a drug: absorption, distribution, metabolism, and excretion.

Pharmacodynamics describes what the drug does to the body, including its effects at receptors.

Half-life is the time it takes for the plasma concentration of a drug to decrease by half.

Drugs with a narrow therapeutic index, such as digoxin, require careful monitoring of blood levels.

Antihypertensives

ACE inhibitors such as lisinopril block the conversion of angiotensin I to angiotensin II, which lowers blood pressure.

A dry, persistent cough is a common side effect of ACE inhibitors.

ACE inhibitors can raise potassium levels, so monitor patients for hyperkalemia.

Calcium channel blockers such as amlodipine relax vascular smooth muscle, which lowers blood pressure.

Cardiac Drugs

Beta blockers such as metoprolol block beta-1 receptors in the heart, which decreases heart rate and lowers blood pressure.

Before giving metoprolol, check the apical pulse; hold the dose and notify the provider if it is below 60 beats per minute.

Digoxin increases the force of contraction of the heart and slows the heart rate.

Low potassium (hypokalemia) increases the risk of digoxin toxicity, so check potassium levels before giving cardiac drugs like digoxin.

Early signs of digoxin toxicity include nausea, vomiting, and visual changes such as yellow-green halos.

Nitroglycerin dilates blood vessels, which reduces the workload of the heart and relieves angina chest pain.

Loop diuretics such as furosemide can cause hypokalemia, which matters for patients who also take digoxin.